SQL

# SQL Interview Pack

8 patterns behind most SQL interview questions. Learn the pattern once, solve 50 questions with it.

- **8 patterns with ready queries**
- **The JOIN mistake that doubles rows**
- **Order of execution, explained**

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# Patterns 1 to 4

Window functions. If you know these, you clear most product and service company rounds.

## Top N per group

Top 3 salaries in each department. DENSE_RANK keeps ties.

```
DENSE_RANK() OVER (PARTITION BY dept
  ORDER BY salary DESC) AS rk  -- then
WHERE rk <= 3
```

## Running total

Cumulative revenue by date.

```
SUM(amount) OVER (ORDER BY order_date)
```

## Month on month change

Compare each row with the previous one.

```
revenue - LAG(revenue) OVER (ORDER BY
month)
```

## Remove duplicates

Keep only the latest row per email.

```
ROW_NUMBER() OVER (PARTITION BY email
  ORDER BY created_at DESC)  -- keep rn =
1
```

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# Patterns 5 to 8

Joins and conditional logic. These separate a fresher answer from an experienced one.

## Consecutive days

Users who logged in 3 days in a row. Same streak = same group value.

```
login_date - ROW_NUMBER() OVER (
  PARTITION BY user_id ORDER BY
login_date)
```

## Find what is missing

Customers who never ordered.

```
LEFT JOIN orders o ON o.cust_id = c.id
WHERE o.id IS NULL
```

## Pivot without PIVOT

Turn status rows into columns.

```
SUM(CASE WHEN status='Returned' THEN 1
ELSE 0 END)
```

## Self join

Employees who earn more than their manager.

```
JOIN employees m ON e.manager_id = m.id
WHERE e.salary > m.salary
```

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# Your JOIN doubled the rows

Most asked debugging question. The answer is almost always duplicate keys.

```
-- Step 1: find the duplicate keys
SELECT customer_id, COUNT(*)
FROM orders
GROUP BY customer_id
HAVING COUNT(*) > 1;
```

## Why it happens

1 row on the left matches 3 rows on the right, so you get 3 rows back. Sums get inflated.

## Fix 1: aggregate first

GROUP BY on the right table, then join.

## Fix 2: deduplicate

Use ROW_NUMBER() and keep rn = 1 before joining.

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# Order of execution

SQL does not run top to bottom. Knowing this answers 5 common questions at once.

```
1 FROM / JOIN      5 SELECT
2 WHERE            6 DISTINCT
3 GROUP BY         7 ORDER BY
4 HAVING           8 LIMIT
```

## Alias fails in WHERE

SELECT runs after WHERE, so the alias does not exist yet.

## WHERE vs HAVING

WHERE filters rows before grouping. HAVING filters groups after.

## Alias works in ORDER BY

ORDER BY runs after SELECT.

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